Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/02/2005
PI Name:	Sivathanu, Yudaya R. Ph.D.		
Project Title:	Fan Beam Emission Tomography for Non-Symmetric Laminar Fires		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47906	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	02/01/2001	**End Date:**	01/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feikema, Douglas A. (Microgravity Science Division, NASA Glenn Research Center) Lim, Jongmook (Combustion Science Branch, En'Urga Inc.)		
Grant/Contract No.:	NAS3-01085		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The four separate global tasks have to be completed to meet the technical objectives of the proposed investigation are: (1) System design and integration of the Fan Beam Emission Tomography (FBET) system for use in the reduced gravity facilities at NASA Glenn Research Center (GRC) including the 2.2 second and 5.0 second drop towers and possibly the KC-135 parabolic aircraft, (2) Development of the deconvolution algorithm and demonstration with synthetic data and a normal gravity test fire, (3) Experimental evaluation of the Fan Beam Emission Tomography System in the reduced gravity facilities at NASA Glenn Research Center, and (4) Preparation of final report and Science Requirements Document (SRD) for hardware and software for use on International Space Station (ISS). The objectives of the research are to (1) develop instrumentation for obtaining emission signatures at multiple view angles from a non-symmetric and transient laminar fire, and (2) develop a deconvolution algorithm to obtain the local		

Task Description:	gas species and soot concentrations and temperature from the emission signatures. It is critical that advanced optical instrumentation be developed for microgravity combustion research on the ISS. This proposed instrument, which enables measurement of temperature and species in a plane, holds great promise to be widely used and important instrument for space-based research. The project involves developing an entire system that is flight worthy, including hardware and analysis software, and then to deliver the system to NASA for use by other researchers. The research involves conducting demonstration experiments in the drop tower at NASA Glenn Research Center with SAL, TIGER-3D, and LSP.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The diagnostics developed for this project is currently used to study rocket engine combustors.
Task Progress:	The first task for the project was to review fires for which microgravity data was available so that the proposed FBET system could be evaluated. Based on the literature survey of quantitative measurements available in normal and reduced gravity experiments, it was decided to evaluate the FBET system using laminar diffusion flames burning ethylene, methane, acetylene, and mixture of these fuels with nitrogen and hydrogen. The second task of the project was to design and fabricate optical scanners to be used in the FBET system. The fabrication of an optical scanner using a computer controlled stepper motor was completed in the fourth month of the project, and the spectrometer with scanner was displayed at the Sixth International Microgravity Combustion Workshop in Cleveland. The scanner provides the path-integrated intensity at 30 angles in the plane of the optical axis. Data was collected at three different heights above the burner for these flames. The data acquisition system logged the data continuously for a period of thirty seconds. The ES100 spectrometer and scanner were successful in obtaining data required for emission tomography. The third task of the project was to procure and integrate the spectrometer and scanner into a drop rig. The spectrometer and scanner was delivered to NASA where it was integrated into a drop rig. In addition, a four channel thermocouple tree for obtaining temperature measurements from the laminar diffusion flames in the drop rig was designed and integrated into the drop rig by NASA. A total of 73 drop tests were conducted during the second year. 31 of these drop tests were conducted with the spectrometer, and 42 drop tests with the thermocouple. The fourth task was to obtain data from a GRC provided combustion chamber at En'Urga Inc. This task was designed to calibrate the spectrometer with 8 mm thick sapphire windows that are currently used in the ISS combustion module. The chamber is not required to calibrate the FBET system with an 8 mm thick sapphire window. Therefore, En'Urga Inc. utilized only the windows and a Santoro type burner (without the combustion chamber) during the second year of the project. The intensities measured from flames were to be corrected for transmission of the windows. This was successfully accomplished during the second year. The fifth task of the project was to review new detectors that could serve as a replacement for the 160 element PbSe array used in the spectrometer. A TE cooled HgCdTe array, manufactured by Sofrad-IR was found to be the only alternative to the current system. However, the HgCdTe array is not available any longer. Therefore, the ES100 with scanner currently offers the best hope of obtaining spectral images from transient flames. The sixth task of the project to develop the deconvolution algorithm required for the FBET system. Forward calculations of the equation of radiative transfer were completed during the first year to obtain synthetic data to be used for the deconvolution algorithm. Deconvolution of the synthetic data so as to obtain temperature and gas species concentrations was completed during the second year of the project. The deconvolution recovered temperatures and gas concentrations with 0.1%. Addition of random noise of 1% in intensity and 10 nm in wavelengths was introduced into the synthetic data. The deconvolution algorithm still allowed for recovery of gas temperatures and gas concentrations within 1% at most locations in the flame. At the center of the flame, the error in the deconvolution procedure was approximately 20%. The seventh task of the project was to obtain data from axisymmetric flames under normal gravity. Normal gravity experiments were conducted at En'Urga Inc. The spectral radiation intensities obtained from the normal gravity experiments were deconvoluted to provide local temperatures, gas concentrations, and soot volume fractions in methane and ethylene flames. For non-sooting flames, reasonable estimates of temperatures and gas concentrations were obtained using the FBET system. For sooting ethylene flames, reasonable estimates of temperatures and soot volume fractions were obtained. However, the gas concentrations could not be retrieved from the deconvolution procedure due to the masking effects of the soot layer. The eighth task of the project was to obtain data from axisymmetric flames under reduced gravity using the 2-second and 5-second drop towers. Two second drop tower experiments were conducted during the third year. Reasonable estimates of temperatures and gas concentrations were obtained from the drop tower experiments for a non-sooting methane flame. For sooting flames, the time required for obtaining data with very high signal to noise ratio was too short. Therefore, it was not possible to obtain reasonable estimates of either temperature or soot volume fractions from these measurements. It was decided to utilize the five second drop tower to obtain data with greater SNR. Five second drop tower tests were conducted and local temperatures, gas concentrations, and soot volume fractions were estimated using the FBET system. The system worked successfully in providing local gas concentrations, temperatures, and soot volume fractions in a non-intrusive manner from microgravity flames for the first time ever.
Bibliography Type:	Description: (Last Updated: 01/07/2020)
Articles in Peer-reviewed Journals	Lim J, Sivathanu Y, Gore J, Li J. "Estimating scalars from spectral radiation measurements in a homogeneous hot gas layer." Combustion and Flame. 2004 Apr;137(1-2):222-9. http://dx.doi.org/10.1016/j.combustflame.2004.02.005 , Apr-2004

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Papers from Meeting Proceedings	Sivathanu Y, Lim J, Feikema D. "Fan Beam Emission Tomography for Laminar Fires." Seventh International Microgravity Workshop, May 2003. Proceedings of the Seventh International Microgravity Workshop, NASA/CP-2003-212376, p. 237-240, May 2003. , May-2003
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